

BG77xA-GL&BG95xA-GL

QCFG AT Commands Manual

LPWA Module Series

Version: 1.0

Date: 2022-05-31

Status: Released



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About the Document

Revision History

Version	Date	Author	Description
-	2021-01-29	Igor STANCIC/ Aleksandar PAUNOVIC/ Mladen VUKOVIC	Creation of the document
1.0	2022-05-31	Igor STANCIC/ Aleksandar PAUNOVIC/ Mladen VUKOVIC	First official release

Contents

About the Document.....	3
Contents	4
Table Index.....	6
1 Introduction	7
1.1. Definitions	7
1.2. AT Command Syntax	7
1.3. AT Command Responses	8
1.4. Declaration of AT Command Examples	9
2 Description of AT+QCFG Commands.....	10
2.1. AT+QCFG Extended Configuration Settings	10
2.1.1. Network-Related AT Commands.....	11
2.1.1.1. AT+QCFG="nwscanmode" Configure RAT(s) to be Searched for.....	11
2.1.1.2. AT+QCFG="servicedomain" Configure Service Domain	12
2.1.1.3. AT+QCFG="nwscanseq" Configure RATs Searching Sequence	13
2.1.1.4. AT+QCFG="band" Configure Frequency Band.....	14
2.1.1.5. AT+QCFG="iotopmode" Configure Network Category to be Searched Under LTE RAT	16
2.1.1.6. AT+QCFG="celevel" Query NB-IoT Coverage Enhancement Level	17
2.1.1.7. AT+QCFG="psm/enter" Trigger the Module into PSM Immediately	18
2.1.1.8. AT+QCFG="psm/urc" Enable/Disable PSM Entering Indication.....	19
2.1.1.9. AT+QCFG="lapiconf" Set LAPI Value (Low Access Priority Indicator)	20
2.1.2. Platform-Related AT Commands	21
2.1.2.1. AT+QCFG="urc/ri/ring" Configure MAIN_RI Behavior in Case of RING URC	21
2.1.2.2. AT+QCFG="urc/ri/smsincoming" Configure MAIN_RI Behavior in Case of Incoming SMS URCs	22
2.1.2.3. AT+QCFG="urc/ri/other" Configure MAIN_RI Behavior in Case of Other URCs.....	23
2.1.2.4. AT+QCFG="risignalttype" Configure MAIN_RI Signal Output Carrier	24
2.1.2.5. AT+QCFG="urc/delay" Configure URC Output Time.....	25
2.1.2.6. AT+QCFG="gpio" Configure GPIO Status	26
2.1.2.7. AT+QCFG="airplanecontrol" Enable/Disable Airplane Mode Control via W_DISABLE#.....	28
2.1.2.8. AT+QCFG="emux/urcport" Configure URC Output Port in EMUX Mode	30
2.1.2.9. AT+QCFG="apready"* Configure AP_READY* Behavior.....	30
2.1.2.10. AT+QCFG="apn/display" Enable/Disable Showing APN	32
2.1.2.11. AT+QCFG="lwm2m" Enable/Disable LwM2M	33
2.1.2.12. AT+QCFG="nwoper" Configure Network Operator Mode of the Modem	33
2.1.2.13. AT+QCFG="ps_dev_mob_type" Configure Device Mobility Type Flag	34
2.1.2.14. AT+QCFG="netupd" Enable/Disable Network Override for Specified LTE Parameters	35
2.1.2.15. AT+QCFG="edrxusimact" Activate/Deactivate (U)SIM in eDRX Mode	37

2.1.2.16.	AT+QCFG="setcfg" Set Configuration Field to the NV memory	38
2.1.2.17.	AT+QCFG="setsyscfg" Set Value(s) to System Configuration Files.....	39
2.1.2.18.	AT+QCFG="uartmapping" Show Port and Function Mapping	40
2.1.2.19.	AT+QCFG="uartmapping/profile" Read/Write Port Profile Mapping Configuration	41
2.1.2.20.	AT+QCFG="usb" Activate/Deactivate USB Interface	42
2.1.2.21.	AT+QCFG="usb/urcport" Configure URC Output Port in USB Interface Mode43	
2.1.2.22.	AT+QCFG="lte/bandprior" Configure Band Scan Priority under LTE	44
2.1.2.23.	AT+QCFG="stored/cell" Delete Stored Cell Acquisition DB	45
3	Summary of +CME ERROR Codes	46
4	Appendix References	48

Table Index

Table 1: Types of AT Commands	8
Table 2: Summary of +CME ERROR Codes	46
Table 3: Related Document.....	48
Table 4: Terms and Abbreviations	48

1 Introduction

This document describes the **AT+QCFG** commands supported on Quectel BG770A-GL, BG772A-GL, BG950A-GL, BG951A-GL, BG952A-GL and BG955A-GL modules.

1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

AT+QCFG commands implemented by the module are in "Extended" syntax, as illustrated below.

- **Extended**

Extended commands can be executed in several modes, as shown in the following table:

Table 1: Types of AT Commands

Command Type	Syntax	Description
Test Command	AT+<cmd>=?	Test the existence of the corresponding command and return information about the type, value, or range of its parameter.
Read Command	AT+<cmd>?	Check the current parameter value of the corresponding command.
Write Command	AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]	Set user-definable parameter value.
Execution Command	AT+<cmd>	Return a specific information parameter or perform a specific action.

Multiple commands can be placed on a single line using a semi-colon (;) between commands. In such cases, only the first command should have **AT** prefix. Commands can be in upper or lower case.

Spaces should be ignored when you enter AT commands, except in the following cases:

- Within quoted strings, where spaces are preserved;
- Within an unquoted string or a numeric parameter;
- Within an IP address;
- Within the AT command name up to and including a =, ? or =?.

On input, at least a carriage return is required. A newline character is ignored so it is permissible to use carriage return/line feed pairs on the input.

If no command is entered after the **AT** token, **OK** will be returned. If an invalid command is entered, **ERROR** will be returned.

Optional parameters, unless explicitly stated, need to be provided up to the last entered parameter.

1.3. AT Command Responses

When the AT command processor has finished processing a line, it will output **OK**, **ERROR** or **+CME ERROR: <err>** to indicate that it is ready to accept a new command. Solicited information responses are sent before the final **OK**, **ERROR** or **+CME ERROR: <err>**.

Responses will be in the format of:

```
<CR><LF>+CMD1:<parameters><CR><LF>
<CR><LF>OK<CR><LF>
```

1.4. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about the use of the AT commands introduced herein. The examples, however, should not be taken as Quectel's recommendations or suggestions about how to design a program flow or what status to set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there is a correlation among these examples, or that they should be executed in a given sequence.

2 Description of AT+QCFG Commands

2.1. AT+QCFG Extended Configuration Settings

The following Test Command shows the supported extended configuration settings of the module.

AT+QCFG Extended Configuration Settings	
Test Command AT+QCFG=?	Response +QCFG: "nwscanmode",(list of supported <scan_mode>s),(list of supported <effect>s) +QCFG: "servicedomain",(list of supported <service>s) +QCFG: "nwscanseq",(range of supported <scanseq>s),(list of supported <effect>s) +QCFG: "band",(range of supported <eMTC_bandval>s),(range of supported <NB-IoT_bandval>s),(list of supported <effect>s) +QCFG: "iotopmode",(range of supported <mode>s),(list of supported <effect>s) +QCFG: "celevel",(range of supported <level>s) +QCFG: "urc/ri/ring",(list of supported <typeRI>s),(range of supported <pulse_duration>s),(range of supported <active_duration>s),(range of supported <inactive_duration>s),(list of supported <ring_no_disturbing>s),(range of supported <pulse_count>s) +QCFG: "urc/ri/smsincoming",(list of supported <typeRI>s),(range of supported <pulse_duration>s),(range of supported <pulse_count>s) +QCFG: "urc/ri/other",(list of supported <typeRI>s),(range of supported <pulse_duration>s),(range of supported <pulse_count>s) +QCFG: "risignalttype",(list of supported <RI_signal_type>s) +QCFG: "urc/delay",(list of supported <enable>s) +QCFG: "ledmode",(list of supported <mode>s) +QCFG: "gpio",(range of supported <mode>s),(list of supported <pin>s),(list of supported <dir>s),(range of supported <pull>s),(list of supported <drv>s),(list of supported <val>s),(list of supported <save>s) +QCFG: "airplanecontrol",(list of supported

	<airplane_control>s) +QCFG: "emux/urcport", (range of supported <URC_port>s) +QCFG: "apready", (list of supported <n>s), (list of supported <level>s), (range of supported <internal>s) +QCFG: "psm/enter", (list of supported <mode>s) +QCFG: "psm/urc", (list of supported <enable>s) +QCFG: "apn/display", (list of supported <mode>s) +QCFG: "lwm2m", (list of supported <enable>s) +QCFG: "nwoper", (list of supported <oper_name>s) +QCFG: "ps_dev_mob_type", (list of supported <flag>s) +QCFG: "netupd", (list of supported <cmd>s) , (list of supported <mode>s) +QCFG: "edrxusimact", (range of supported <mode>s) +QCFG: "setcfg", (list of supported <obj>s), (list of supported <mode>s) +QCFG: "setsyscfg", (list of supported <obj>s), (range of supported <stage>s), (list of supported <enable>s), (range of supported <lower_fr>s), (range of supported <upper_fr>s), (range of supported <state>s) +QCFG: "uartmapping" +QCFG: "usb", (list of supported <enable>s) +QCFG: "usb/urcport", (range of supported <URC_port>s) OK
Maximum Response Time	300 ms

2.1.1. Network-Related AT Commands

2.1.1.1. AT+QCFG="nwscanmode" Configure RAT(s) to be Searched for

The command queries and configures the RAT(s) to be searched for.

AT+QCFG="nwscanmode" Configure RAT(s) to be Searched for	
Write Command	Response
AT+QCFG="nwscanmo"[,<scan_mode>,<effect>]]	If the optional parameter is omitted, query the current settings: +QCFG: "nwscanmode",<scan_mode> OK If any of the optional parameter is specified, configure the RAT(s) to be searched for: OK

	If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	<effect> determines when the command will take effect. The configurations are saved automatically.

Parameter

<scan_mode>	Integer type. RAT(s) to be searched for. <u>0</u> Automatic (GSM and LTE) 1 GSM only 3 LTE only
<effect>	Integer type. When to take effect. 0 Take effect after rebooting <u>1</u> Take effect immediately
<err>	Error code. See Chapter 3 for more information.

NOTE

This command is valid only on BG955A-GL module.

2.1.1.2. AT+QCFG="servicedomain" Configure Service Domain

The command queries and configures the service domain to be registered.

AT+QCFG="servicedomain" Configure Service Domain	
Write Command AT+QCFG="servicedomain"[,<service>]	Response If the optional parameter is omitted, query the current settings: +QCFG: "servicedomain",<service> OK If the optional parameter is specified, configure the service domain to be registered: OK If there is an error related to ME functionality:

	+CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The configurations are saved automatically.

Parameter

<service>	Integer type. Service domain. 1 PS only
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.1.3. AT+QCFG="nwscanseq" Configure RATs Searching Sequence

The command queries and configures RATs searching sequence.

AT+QCFG="nwscanseq" Configure RATs Searching Sequence	
Write Command	Response
AT+QCFG="nwscanseq"[,<scanseq>[,effect]]	If optional parameters are omitted, query the current settings: +QCFG: "nwscanseq",<scanseq> OK If any of the optional parameters is specified, configure RATs searching sequence: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	<effect> determines when the command takes effect. The configurations are saved automatically.

Parameter

<scanseq>	Integer type. RATs searching sequence. (e.g.: 020301 stands for eMTC → NB-IoT → GSM) <u>00</u> Automatic (eMTC → NB-IoT) 01 GSM 02 eMTC 03 NB-IoT
<effect>	Integer type. Determines when the command takes effect. 0 Take effect after the module reboots <u>1</u> Take effect immediately
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.1.4. AT+QCFG="band" Configure Frequency Band

The command queries and configures the frequency bands to be searched.

AT+QCFG="band" Configure Frequency Band	
Write Command AT+QCFG="band",<GSM_bandval>,<eMTC_bandval>,<NB-IoT_bandval>[,<effect>]]	Response If optional parameters are omitted, query the current settings: +QCFG: "band",<GSM_bandval>,<eMTC_bandval>,<NB-IoT_bandval> OK If any of the optional parameters is specified, configure the frequency bands to be searched. OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	4500 ms
Characteristics	<effect> determines when the command takes effect. The configurations are saved automatically.

Parameter

<GSM_bandval>	A hexadecimal value that specifies the GSM frequency band (e.g.: 0xa = 0x2(DCS1800) + 0x8(PCS1900)). If it is set to 0, it means not to change GSM frequency band. <table> <tr> <td>0</td><td>No change</td></tr> <tr> <td>0x1</td><td>EGSM900</td></tr> <tr> <td>0x2</td><td>DCS1800</td></tr> <tr> <td>0x4</td><td>GSM850</td></tr> <tr> <td>0x8</td><td>PCS1900</td></tr> <tr> <td>0Xf</td><td>All of the supported bands above</td></tr> </table>	0	No change	0x1	EGSM900	0x2	DCS1800	0x4	GSM850	0x8	PCS1900	0Xf	All of the supported bands above																						
0	No change																																		
0x1	EGSM900																																		
0x2	DCS1800																																		
0x4	GSM850																																		
0x8	PCS1900																																		
0Xf	All of the supported bands above																																		
<eMTC_bandval>	A hexadecimal value that specifies the eMTC frequency band (e.g.: 0x15 = 0x01(LTE B1) + 0x04(LTE B3) + 0x10(LTE B5)). If it is set to 0, the eMTC frequency band will not be changed. By default, all bands are set. <table> <tr> <td>0</td><td>No change</td></tr> <tr> <td>0x1 (BAND_PREF_LTE_BAND1)</td><td>LTE B1</td></tr> <tr> <td>0x2 (BAND_PREF_LTE_BAND2)</td><td>LTE B2</td></tr> <tr> <td>0x4 (BAND_PREF_LTE_BAND3)</td><td>LTE B3</td></tr> <tr> <td>0x8 (BAND_PREF_LTE_BAND4)</td><td>LTE B4</td></tr> <tr> <td>0x10 (BAND_PREF_LTE_BAND5)</td><td>LTE B5</td></tr> <tr> <td>0x80 (BAND_PREF_LTE_BAND8)</td><td>LTE B8</td></tr> <tr> <td>0x800 (BAND_PREF_LTE_BAND12)</td><td>LTE B12</td></tr> <tr> <td>0x1000 (BAND_PREF_LTE_BAND13)</td><td>LTE B13</td></tr> <tr> <td>0x20000 (BAND_PREF_LTE_BAND18)</td><td>LTE B18</td></tr> <tr> <td>0x40000 (BAND_PREF_LTE_BAND19)</td><td>LTE B19</td></tr> <tr> <td>0x80000 (BAND_PREF_LTE_BAND20)</td><td>LTE B20</td></tr> <tr> <td>0x1000000 (BAND_PREF_LTE_BAND25)</td><td>LTE B25</td></tr> <tr> <td>0x2000000 (BAND_PREF_LTE_BAND26)</td><td>LTE B26</td></tr> <tr> <td>0x4000000 (BAND_PREF_LTE_BAND27)</td><td>LTE B27</td></tr> <tr> <td>0x8000000 (BAND_PREF_LTE_BAND28)</td><td>LTE B28</td></tr> <tr> <td>0x2000000000000000 (BAND_PREF_LTE_BAND66)</td><td>LTE B66</td></tr> </table>	0	No change	0x1 (BAND_PREF_LTE_BAND1)	LTE B1	0x2 (BAND_PREF_LTE_BAND2)	LTE B2	0x4 (BAND_PREF_LTE_BAND3)	LTE B3	0x8 (BAND_PREF_LTE_BAND4)	LTE B4	0x10 (BAND_PREF_LTE_BAND5)	LTE B5	0x80 (BAND_PREF_LTE_BAND8)	LTE B8	0x800 (BAND_PREF_LTE_BAND12)	LTE B12	0x1000 (BAND_PREF_LTE_BAND13)	LTE B13	0x20000 (BAND_PREF_LTE_BAND18)	LTE B18	0x40000 (BAND_PREF_LTE_BAND19)	LTE B19	0x80000 (BAND_PREF_LTE_BAND20)	LTE B20	0x1000000 (BAND_PREF_LTE_BAND25)	LTE B25	0x2000000 (BAND_PREF_LTE_BAND26)	LTE B26	0x4000000 (BAND_PREF_LTE_BAND27)	LTE B27	0x8000000 (BAND_PREF_LTE_BAND28)	LTE B28	0x2000000000000000 (BAND_PREF_LTE_BAND66)	LTE B66
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0x4 (BAND_PREF_LTE_BAND3)	LTE B3																																		
0x8 (BAND_PREF_LTE_BAND4)	LTE B4																																		
0x10 (BAND_PREF_LTE_BAND5)	LTE B5																																		
0x80 (BAND_PREF_LTE_BAND8)	LTE B8																																		
0x800 (BAND_PREF_LTE_BAND12)	LTE B12																																		
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0x4000000 (BAND_PREF_LTE_BAND27)	LTE B27																																		
0x8000000 (BAND_PREF_LTE_BAND28)	LTE B28																																		
0x2000000000000000 (BAND_PREF_LTE_BAND66)	LTE B66																																		
<NB-IoT_bandval>	A hexadecimal value that specifies the NB-IoT frequency band (e.g.: 0x15 = 0x01(LTE B1) + 0x04(LTE B3) + 0x10(LTE B5)). If it is set to 0, the NB-IoT frequency band will not be changed. By default, all bands are set. <table> <tr> <td>0</td><td>No change</td></tr> <tr> <td>0x1 (BAND_PREF_LTE_BAND1)</td><td>LTE B1</td></tr> <tr> <td>0x2 (BAND_PREF_LTE_BAND2)</td><td>LTE B2</td></tr> <tr> <td>0x4 (BAND_PREF_LTE_BAND3)</td><td>LTE B3</td></tr> <tr> <td>0x8 (BAND_PREF_LTE_BAND4)</td><td>LTE B4</td></tr> <tr> <td>0x10 (BAND_PREF_LTE_BAND5)</td><td>LTE B5</td></tr> <tr> <td>0x80 (BAND_PREF_LTE_BAND8)</td><td>LTE B8</td></tr> <tr> <td>0x800 (BAND_PREF_LTE_BAND12)</td><td>LTE B12</td></tr> <tr> <td>0x1000 (BAND_PREF_LTE_BAND13)</td><td>LTE B13</td></tr> </table>	0	No change	0x1 (BAND_PREF_LTE_BAND1)	LTE B1	0x2 (BAND_PREF_LTE_BAND2)	LTE B2	0x4 (BAND_PREF_LTE_BAND3)	LTE B3	0x8 (BAND_PREF_LTE_BAND4)	LTE B4	0x10 (BAND_PREF_LTE_BAND5)	LTE B5	0x80 (BAND_PREF_LTE_BAND8)	LTE B8	0x800 (BAND_PREF_LTE_BAND12)	LTE B12	0x1000 (BAND_PREF_LTE_BAND13)	LTE B13																
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0x2 (BAND_PREF_LTE_BAND2)	LTE B2																																		
0x4 (BAND_PREF_LTE_BAND3)	LTE B3																																		
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	0x20000 (BAND_PREF_LTE_BAND18)	LTE B18
	0x40000 (BAND_PREF_LTE_BAND19)	LTE B19
	0x80000 (BAND_PREF_LTE_BAND20)	LTE B20
	0x1000000 (BAND_PREF_LTE_BAND25)	LTE B25
	0x8000000 (BAND_PREF_LTE_BAND28)	LTE B28
	0x2000000000000000 (BAND_PREF_LTE_BAND66)	LTE B66
<effect>	Integer type. Determines when the command takes effect.	
	0	Take effect after the module reboots
	1	Take effect immediately
<err>	Integer type. Error code. See Chapter 3 for more information.	

NOTE

<GSM_bandval> is valid only on BG955A-GL module.

2.1.1.5. AT+QCFG="iotopmode" Configure Network Category to be Searched Under LTE RAT

The command queries and configures the network category to be searched under LTE RAT.

AT+QCFG="iotopmode" Configure Network Category to be Searched Under LTE RAT	
Write Command AT+QCFG="iotopmode"[,<mode>,<effect>]]	Response If optional parameters are omitted, query the current settings: +QCFG: "iotopmode",<mode> OK If any of the optional parameters is specified, configure the network category to be searched under LTE RAT: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	4500 ms
Characteristics	<effect> determines when the command takes effect. The configurations are saved automatically.

Parameter

<mode>	Integer type. Network category to be searched under LTE RAT. 0 eMTC 1 NB-IoT <u>2</u> eMTC and NB-IoT
<effect>	Integer type. Determines when the command takes effect. 0 Take effect after the module reboots <u>1</u> Take effect immediately
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.1.6. AT+QCFG="celevel" Query NB-IoT Coverage Enhancement Level

The command queries the NB-IoT coverage enhancement (CE) level.

AT+QCFG="celevel" Query NB-IoT Coverage Enhancement Level	
Write Command AT+QCFG="celevel"	Response: +QCFG: "celevel",<level> OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	/

Parameter

<level>	Integer type. NB-IoT coverage enhancement level. 0 CE level 0 1 CE level 1 2 CE level 2
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.1.7. AT+QCFG="psm/enter" Trigger the Module into PSM Immediately

The command queries the current settings or sets whether to trigger the module into the Power Saving Mode (PSM) immediately after the RRC connection release is received.

When **<mode>**=1, the module skips the active timer (T3324) and enters PSM immediately after the RRC connection release is received.

AT+QCFG="psm/enter" Trigger the Module into PSM Immediately	
Write Command AT+QCFG="psm/enter"[,<mode>]	Response If the optional parameter is omitted, query the current settings: +QCFG: "psm/enter",<mode> OK If the optional parameter is specified, set whether to trigger the module into PSM immediately after the RRC connection release is received: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<mode>	Integer type. Sets whether to trigger the module into PSM. <u>0</u> Enter PSM after T3324 expires 1 Enter PSM immediately after the RRC connection release is received.
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.1.8. AT+QCFG="psm/urc" Enable/Disable PSM Entering Indication

The command queries the current settings and enables/disables the output of URC **+QPSMTIMER: <TAU_timer>, <T3324_timer>**, which indicates the TAU duration and active time duration of the module's PSM.

When the PSM function is enabled and the RRC connection release is received, the active timer (T3324) is started, and the indication URC is reported.

AT+QCFG="psm/urc" Enable/Disable PSM Entering Indication	
Write Command AT+QCFG="psm/urc",<enable>	Response If the optional parameter is omitted, query the current settings: +QCFG: "psm/urc",<enable> OK If the optional parameter is specified, set whether to enable PSM entering indication. OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<enable>	Integer type. Enable/disable the output of URC +QPSMTIMER: <TAU_timer>, <T3324_timer>. If enabled, the URC will be reported when the RRC connection release is received. <u>0</u> Disable 1 Enable
<err>	Integer type. Error code. See Chapter 3 for more information.

NOTE

When **AT+QCFG="psm/urc",1** and **AT+QCFG="psm/enter",1** are executed at the same time, it is possible that the URC **+QPSMTIMER** cannot be outputted because the module enters PSM right away.

2.1.1.9. AT+QCFG="lapiconf" Set LAPI Value (Low Access Priority Indicator)

The command queries the current settings or configures the Low Access Priority Indicator.

AT+QCFG="lapiconf" Set LAPI Value (Low Access Priority Indicator)	
Write Command AT+QCFG="lapiconf"[,<mode>][,<enable>]	Response If the optional parameter is omitted, query the current settings: +QCFG: "lapiconf",<mode>[,<enable>] OK If the optional parameter is specified, configure the URC output port in USB Interface mode: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The configuration is saved automatically.

Parameter

<mode>	Integer type. Whether to enable low access priority indication feature 0 Disable. Forced to disable 1 Enable. Forced to enable 2 Auto. Determined by EFS file
<enable>	Integer type. 0 Disable 1 Enable
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.2. Platform-Related AT Commands

Among the following AT commands, **AT+QCFG="urc/ri/ring"**, **AT+QCFG="urc/ri/smsincoming"** and **AT+QCFG="urc/ri/other"** commands control the behavior of the MAIN_RI pin when a URC is reported. The MAIN_RI pin is active low.

2.1.2.1. AT+QCFG="urc/ri/ring" Configure MAIN_RI Behavior in Case of RING URC

The command queries and configures the behavior of MAIN_RI pin when **RING** URC is reported to indicate an incoming call.

The sum of **<active_duration>** and **<inactive_duration>** determines the interval of RING indications when a call is coming.

AT+QCFG="urc/ri/ring" Configure MAIN_RI Behavior in Case of RING URC	
Write Command AT+QCFG="urc/ri/ring",<typeRI>,<pulse_duration>,<active_duration>,<inactive_duration>,<ring_no_disturbing>,<pulse_count>]]]]]	Response If optional parameters are omitted, query the current settings: +QCFG: "urc/ri/ring",<typeRI>,<pulse_duration>,<active_duration>,<inactive_duration>,<ring_no_disturbing>,<pulse_count> OK If any of the optional parameters is specified, configure the behavior of MAIN_RI pin when RING URC is reported. OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are saved automatically.

Parameter

<typeRI>	String type. MAIN_RI behavior when RING URCs are reported.
"off"	No change. MAIN_RI pin stays inactive (high).
"pulse"	Pulse. Pulse width is determined by <pulse_duration> .

	"always"	Change to active. MAIN_RI behavior can be restored to inactive with AT+QIR (see document [1] for more information).
	"auto"	When RING URC is reported to indicate an incoming call, MAIN_RI pin changes to active and remains active. Answering or hanging up the incoming call changes MAIN_RI pin state to inactive.
	"wave"	When RING URC is reported to indicate an incoming call, MAIN_RI pin outputs a square wave. Both <active_duration> and <inactive_duration> are used to set the square wave. Answering or hanging up the incoming call changes the MAIN_RI pin to inactive.
<pulse_duration>	Integer type. Pulse width. Range: 1–2000. Default: 120. Unit: ms. Parameter is valid only when <typeRI>="pulse" .	
<active_duration>	Integer type. Active duration of the square wave. Range: 1–10000. Default: 1000. Unit: ms. Parameter is valid only when <typeRI>="wave" .	
<inactive_duration>	Integer type. Inactive duration of the square wave. Range: 1–10000. Default: 5000. Unit: ms. Parameter is valid only when <typeRI>="wave" .	
<ring_no_disturbing>	String type. Set whether the MAIN_RI pin behavior could be disturbed. Parameter is valid only when <typeRI>="auto" or "wave" . For example, when <typeRI>="wave" , if the square wave will not be interrupted by other URCs (including SMS-related URCs), then <ring_no_disturbing> should be set to "on" . "off" MAIN_RI pin behavior can be changed by other URCs when an incoming call is ringing. "on" MAIN_RI behavior cannot be changed by other URCs when an incoming call is ringing.	
<pulse_count>	Integer type. Pulse count. Parameter is valid only when <typeRI>="pulse" . Range: 1–5. Default: 1. Interval between two pulses is equal to <pulse_duration> .	
<err>	Integer type. Error code. See Chapter 3 for more information.	

2.1.2.2. AT+QCFG="urc/ri/smsincoming" Configure MAIN_RI Behavior in Case of Incoming SMS URCs

The command queries and configures the behavior of the MAIN_RI pin when there are incoming SMS URCs. Incoming SMS URCs include **+CMTI**, **+CMT**, **+CDS**, and **+CBM** (see **document [1]** for details).

AT+QCFG="urc/ri/smsincoming" Configure MAIN_RI Behavior in Case of Incoming SMS URCs	
Write Command AT+QCFG="urc/ri/smsincoming"[,<typeRI>,<pulse_duration>,<pulse_count>]]]	Response If optional parameters are omitted, query the current settings: +QCFG: "urc/ri/smsincoming",<typeRI>,<pulse_duration>

	n>,<pulse_count> OK If any of the optional parameters is specified, configure the MAIN_RI behavior when there are incoming SMS URCs: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are saved automatically.

Parameter

<typeRI>	String type. MAIN_RI behavior when there are incoming SMS URCs. "off" No change. MAIN_RI pin stays inactive (high). "pulse" Pulse. Pulse width is determined by <pulse_duration>. "always" Change to active. MAIN_RI behavior can be restored to inactive with AT+QIR (see document [1] for more information).
<pulse_duration>	Integer type. Pulse width. Range: 1–2000. Default: 120. Unit: ms. Parameter is valid only when <typeRI>="pulse".
<pulse_count>	Integer type. Pulse count. Parameter is valid only when <typeRI>="pulse". Range: 1–5. Default: 1. Interval between two pulses is equal to <pulse_duration>.
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.2.3. AT+QCFG="urc/ri/other" Configure MAIN_RI Behavior in Case of Other URCs

The command queries and configures the behavior of the MAIN_RI pin when there are other URCs.

AT+QCFG="urc/ri/other" Configure MAIN_RI Behavior in Case of Other URCs	
Write Command	Response
AT+QCFG="urc/ri/other"[,<typeRI>[,<pulse_duration>[,<pulse_count>]]]	If optional parameters are omitted, query the current settings: +QCFG: "urc/ri/other",<typeRI>,<pulse_duration>,<pulse_count>

	OK If any of the optional parameters is specified, configure the MAIN_RI behavior when there are other URCs. OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are saved automatically.

Parameter

<typeRI>	String type. MAIN_RI behavior when there are other URCs. "off" No change. MAIN_RI pin stays inactive (high). "pulse" Pulse. Pulse width determined by <pulse_duration> .
<pulse_duration>	Integer type. Pulse width. Range: 1–2000. Default: 120. Unit: ms. Parameter is valid only when <typeRI> = "pulse".
<pulse_count>	Integer type. Pulse count. Parameter is valid only when <typeRI> = "pulse". Range: 1–5. Default: 1. Interval between two pulses is equal to <pulse_duration> .
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.2.4. AT+QCFG="risignalttype" Configure MAIN_RI Signal Output Carrier

The command queries and configures the output carrier of the MAIN_RI signal.

AT+QCFG="risignalttype" Configure Main_RI Signal Output Carrier	
Write Command AT+QCFG="risignalttype"[,<RI_signal_type>]	Response If the optional parameter is omitted, query the current settings: +QCFG: "risignalttype",<RI_signal_type> OK If the optional parameter is specified, configure the output carrier of the MAIN_RI signal.

	OK If there is an error related to the ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<RI_signal_type>	String type. MAIN_RI signal output carrier.
"respective"	Causes MAIN_RI behavior on the port where a URC is presented. For example, if a URC is presented on the UART port, it is a physical ring indication signal. AT+QURCCFG="urcport" can get the port on which a URC is presented (see document [1] for more information).
"physical"	No matter on which port the URC is presented, it only causes the behavior of physical ring indication signal.
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.2.5. AT+QCFG="urc/delay" Configure URC Output Time

The command queries and configures URC output time.

AT+QCFG="urc/delay" Configure URC Output Time	
Write Command AT+QCFG="urc/delay"[,<enable>]	Response If the optional parameter is omitted, query the current settings: +QCFG: "urc/delay",<enable> OK If the optional parameter is specified, configure URC output time: OK If there is an error related to ME functionality: +CME ERROR: <err>

	If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<enable>	Integer type. URC output time. 0 Output URC when ring indication pulse starts. 1 Output URC when ring indication pulse ends (effective only when <typeRI>="pulse" . See AT+QCFG="urc/ri/ring" , AT+QCFG="urc/ri/smsincoming" and AT+QCFG="urc/ri/other" for more information).
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.2.6. AT+QCFG="gpio" Configure GPIO Status

The command queries and configures the GPIO status.

AT+QCFG="gpio" Configure GPIO Status	
Write Command Query the formats of the command AT+QCFG="gpio"	Response +QCFG: "gpio",<mode>,<pin>[,<dir>,<pull>,<drv>]/[<val>][,<save>]] OK
Write Command AT+QCFG="gpio",<mode>,<pin>[,<dir>,<pull>,<drv>]/[<val>][,<save>]]	Response If <mode>=2 , then all optional parameters should be omitted. +QCFG: "gpio",<val> OK If <mode>=1 , then <val> should be omitted. OK If <mode>=3 , then <dir> , <pull> and <drv> should be omitted. OK If there is an error related to ME functionality: +CME ERROR: <err>
Maximum Response Time	300 ms

Characteristics

The command takes effect immediately.

<save> determines whether the configuration will be saved. When **<save>=1**, the configurations are saved and remain valid after the module reboots.

Parameter

<mode> Integer type. Command mode.

- 1 Initialize GPIO status
- 2 Query GPIO status
- 3 Configure GPIO status

<pin> Integer type. GPIO pin number.

BG770A module supports the following pin numbers:

Pin No.	Pin Name
1	GPIO1
8	GPIO2
9	GPIO3
33	GPIO4
40	GPIO5
57	GPIO6
63	GPIO7

BG950A-GL, BG951A-GL and BG955A-GL modules support the following pin numbers:

Pin No.	Pin Name
25	GPIO1
26	GPIO2
64	GPIO3
65	GPIO4
66	GPIO5
85	GPIO6
86	GPIO7
87	GPIO8
88	GPIO9

<dir> Integer type. GPIO pin direction.

- 0 Input
- 1 Output

<pull> Integer type. GPIO pin pull type.

- 0 No pull
- 1 Pull the GPIO down
- 2 Keep the GPIO as it is
- 3 Pull the GPIO up

<drv> Integer type. GPIO pin drive strength.

- 0 2 mA
- 1 4 mA
- 3 8 mA

	5 12 mA
<val>	Integer type. Value read from or written to a GPIO.
	0 Low level
	1 High level
<save>	Integer type. Determines whether to save the configurations.
	0 Do not save
	1 Save
<err>	Integer type. Error code. See Chapter 3 for more information.

NOTE

1. Command **AT+QCFG="gpio"** cannot be used on BG772A-GL and BG952A-GL modules.
2. **<save>** is valid only when **<mode>** is 1 or 3.
3. For the value of **<pin>**, see corresponding hardware design for details.

2.1.2.7. AT+QCFG="airplanecontrol" Enable/Disable Airplane Mode Control via W_DISABLE#

The command enables/disables airplane mode control via the W_DISABLE# pin and queries the current settings. If the function is enabled, the module enters the airplane mode when the pin is pulled down, or the normal mode when the pin is pulled up. In addition, the command outputs a related URC before the module enters or exits airplane mode.

AT+QCFG="airplanecontrol" Enable/Disable Airplane Mode Control via W_DISABLE#	
Write Command AT+QCFG="airplanecontrol",<airplane_control>]	Response If the optional parameter is omitted, query the current settings: +QCFG: "airplanecontrol",<airplane_control>,<airplane_status> OK If the optional parameter is specified, configure whether to enable airplane mode control via W_DISABLE# pin: OK Or ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<airplane_control>	Integer type. Enable/disable airplane mode control via W_DISABLE# pin. 0 Disable 1 Enable
<airplane_status>	Integer type. Current status. 0 In normal mode 1 In airplane mode

NOTE

The status of the W_DISABLE# pin may affect the validity of **AT+CFUN** (see **document [1]** for more information). When the airplane mode control via W_DISABLE# is enabled and the pin is pulled down, the module enters the airplane mode regardless of the status of **AT+CFUN**. In addition, the module functionality level cannot be switched with **AT+CFUN**.

Example

```

AT+QCFG="airplanecontrol",1 //Enable airplane mode control via W_DISABLE#.
OK
//Pull down W_DISABLE# pin
+QIND: "airplanestatus",1 //URC indicating that the module enters airplane mode.

AT+QCFG="airplanecontrol" //Query whether airplane mode control via W_DISABLE# is enabled
and the current status of the module.
+QCFG: "airplanecontrol",1,1 //Airplane mode control via W_DISABLE# is enabled and the module
is currently in airplane mode.

OK
//Pull up W_DISABLE# pin
+QIND: "airplanestatus",0 //URC indicating that the module exits airplane mode.

AT+QCFG="airplanecontrol" //Query whether airplane mode control via W_DISABLE# is enabled
and the current status of the module.
+QCFG: "airplanecontrol",1,0 //Airplane mode control via W_DISABLE# is enabled and the module
is currently in normal mode.

OK

```

2.1.2.8. AT+QCFG="emux/urcport" Configure URC Output Port in EMUX Mode

The command queries and configures the URC output port in EMUX mode.

AT+QCFG="emux/urcport" Configure URC Output Port in EMUX Mode	
Write Command AT+QCFG="emux/urcport"[,<URC_port>]	Response If the optional parameter is omitted, query the current settings: +QCFG: "emux/urcport",<URC_port> OK If the optional parameter is specified, configure the URC output port in EMUX mode: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<URC_port>	Integer type. URC output port in EMUX mode. 0 All ports 1 Virtual port 1 2 Virtual port 2 3 Virtual port 3
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.2.9. AT+QCFG="apready"* Configure AP_READY* Behavior

The command queries and configures the behavior of the AP_READY pin. An external MCU can change AP_READY pin level as needed.

When there is a URC to be reported, if the AP_READY pin level is invalid, the URC is buffered first, and the AP_READY pin level will be detected periodically within the configured detection period. The URC will

be output when the AP_READY pin level becomes valid. The pulse signal generated on the MAIN_RI pin can still be output according to the configured mode, and will not be buffered.

AT+QCFG="apready" Configure AP_READY Behavior

Write Command AT+QCFG="apready",<n>[,<level>[,<interval>]]]	Response If the optional parameters are omitted, query the current settings: +QCFG: "apready",<n>,<level>,<interval> OK If any of the optional parameters is specified, configure the AP_READY behavior: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are saved automatically.

Parameter

<n>	Integer type. Enable/disable the AP_READY pin for AP sleep state detection. 0 Disable 1 Enable
<level>	Integer type. Valid level of AP_READY pin. Parameter is valid only when the AP_READY detection function is enabled. 0 Low level 1 High level
<interval>	Integer type. Detection interval. Range: 100–3000. Default: 500. Unit: ms. Parameter is valid only when the AP_READY detection function is enabled.
<err>	Integer type. Error code. See Chapter 3 for more information.

NOTE

- Up to 15 URCs can be buffered. When the number of URCs exceeds 15, the oldest one in the buffer will be cleared to store a new URC.
- The **RING** URC is buffered only once for each call process.

2.1.2.10. AT+QCFG="apn/display" Enable/Disable Showing APN

The command enables/disables showing of allocated point name (APN) and queries the current settings. When **<mode>=1**, the APN is shown in the response of **AT+CGDCONT?** (see **document [1]** for more information).

AT+QCFG="apn/display" Enable/Disable Showing APN	
Write Command AT+QCFG="apn/display",<mode>	Response If the optional parameter is omitted, query the current settings: +QCFG: "apn/display",<mode> OK If the optional parameter is specified, set whether to enable showing APN. OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<mode>	Integer type. Enable/disable showing APN. 0 Disable 1 Enable
<err>	Integer type. Error code. Refer to Chapter 3 for more information.

2.1.2.11. AT+QCFG="lwm2m" Enable/Disable LwM2M

The command enables/disables LwM2M and queries the current settings.

AT+QCFG="lwm2m" Enable/Disable LwM2M	
Write Command AT+QCFG="lwm2m"[,<enable>]	Response If the optional parameter is omitted, query the current settings: +QCFG: "lwm2m",<enable> OK If the optional parameter is specified, set whether to enable LwM2M. OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect after the module reboots. The configuration is saved automatically.

Parameter

<enable>	Integer type. 0 Disable 1 Enable
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.2.12. AT+QCFG="nwoper" Configure Network Operator Mode of the Modem

The command queries and configures network operator mode of the modem. This mode setting supports network operator specific requirements defined on top of 3GPP requirements. **<oper_name>="DEFAULT"** means default 3GPP compliant behavior of the modem. The list of operators is not limited. Use Test Command (**AT+QCFG=?**) to retrieve the list of currently supported operators. Any attempt to set unknown operator name will return **ERROR**.

In the "AUTO" mode, when a new operator is identified, the configurations are set by the modem similar to

AT+QCFG="nwoper",<oper_name> operation. This means all operator configuration files will be restored to the new selected operator default values.

AT+QCFG="nwoper" Configure Network Operator Mode of the Modem	
Write Command AT+QCFG="nwoper" [<oper_name>]	Response If the optional parameter is omitted, query the current settings: +QCFG: "nwoper",<oper_name>[,mode] OK If the optional parameter is specified, set network operator mode: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<oper_name>	String type. Network operator mode of the modem. <u>"DEFAULT"</u> Default 3GPP compliant mode "VZW" Verizon "ATT" AT&T ...
<mode>	String type. Optional parameter, indicates that "AUTO" mode is currently configured. <u>"AUTO"</u> Network operator mode is selected in accordance with IMSI value of the currently used (U)SIM
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.2.13. AT+QCFG="ps_dev_mob_type" Configure Device Mobility Type Flag

The command queries and configures device mobility type flag. It gets configuration from JSON configuration file.

AT+QCFG="ps_dev_mob_type" Configure Device Mobility Type Flag	
Write Command AT+QCFG="ps_dev_mob_type",<flag>]	Response If the optional parameter is omitted, query the current settings: +QCFG: "ps_dev_mob_type",<flag> OK If the optional parameter is specified, set device mobility type flag: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<flag>	Integer type. Device mobility type flag. 1 mobile 2 static
<err>	Integer type. Error code. See Chapter 3 for more information.

Example

AT+QCFG="ps_dev_mob_type",1 OK	//Set the flag to mobile.
---	---------------------------

2.1.2.14. AT+QCFG="netupd" Enable/Disable Network Override for Specified LTE Parameters

The command queries the current settings and enables/disables network override for specified LTE parameters (i.e., by EMM messages).

AT+QCFG="netupd" Enable/Disable Network Override for Specified LTE Parameters

Write Command AT+QCFG="netupd",<cmd>[,<mode>]	Response If the optional parameter is omitted, query the current settings: +QCFG: "netupd",<cmd>,<mode> OK If the optional parameter is specified, enable/disable network override for specified LTE parameters: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are not saved.

Parameter

<cmd>	String type. "NWNNAME" Sets the behavior of network name supplied by AT+COPS? .
<mode>	Integer type. For "NWNNAME": <u>0</u> AT+COPS? show the most updated full network name as required by the 27.007 standard. 1 Prohibit override of network name by EMM message (i.e. shown in AT+COPS? , etc.) even if the EMM information message indicates another full network name.
<err>	Integer type. Error code. See Chapter 3 for more information.

Example

```
AT+QCFG="netupd","nwname",1 //Set "nwname" to mode 1.
OK
```

2.1.2.15. AT+QCFG="edrxusimact" Activate/Deactivate (U)SIM in eDRX Mode

This command activates/deactivates (U)SIM in eDRX mode regardless of (U)SIM EF_AD settings.

AT+QCFG="edrxusimact" Activate/Deactivate (U)SIM in eDRX Mode	
Write Command AT+QCFG="edrxusimact"[,<mode>]	Response If the optional parameter is omitted, query the current settings: +QCFG: "edrxusimact",<mode> OK If the optional parameter is specified, activate/deactivate (U)SIM in eDRX mode regardless of (U)SIM EF_AD settings: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<mode>	Integer type. (U)SIM activation policy. <u>0</u> Apply default (U)SIM activation policy as per (U)SIM EF_AD settings in eDRX mode 1 Activate (U)SIM in eDRX mode regardless of (U)SIM EF_AD settings 2 Deactivate (U)SIM in eDRX mode regardless of (U)SIM EF_AD settings
<err>	Integer type. Error code. See Chapter 3 for more information.

Example

AT+QCFG="edrxusimact"	//Query the current (U)SIM activation policy.
+QCFG: "edrxusimact",0	
OK	

2.1.2.16. AT+QCFG="setcfg" Set Configuration Field to the NV memory

This command sets configuration field to the NV memory.

AT+QCFG="setcfg" Set Configuration Field to the NV memory	
Write Command AT+QCFG="setcfg",<obj>[,<mode>]	Response If the optional parameter is omitted, query the current settings: +QCFG: <obj>,<mode> OK If the optional parameter is specified, activate/deactivate USB interface: OK If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are saved automatically.

Parameter

<obj>	String type.	
	"sim_dual_config"	Enable the dual SIM feature (between plastic and embedded SIM only)
	"sim_init_select_policy"	SIM policy options
	"data_inactivity_dis"	Enable/Disable inactivity timer
<mode>	Integer type.	
	For "sim_dual_config":	
	0 SW default	
	1 Single SIM	
	2 Dual SIM	
	For "sim_init_select_policy":	
	0 N/A, Single SIM	
	1 SIM1 only	
	2 Reserved	
	3 Reserved	
	4 Reserved	
	5 iUICC only option (no fallback available)	
	For "data_inactivity_dis":	

- 0 Inactivity timer is enabled
- 1 inactivity timer is disabled

Example

```
AT+QCFG="setcfg"," sim_init_select_policy ",5 // set to iSIM only
OK
```

2.1.2.17. AT+QCFG="setsyscfg" Set Value(s) to System Configuration Files

This command sets value(s) to system configuration files.

AT+QCFG="setsyscfg" Set Value(s) to System Configuration Files	
Write Command AT+QCFG="setsyscfg",<obj>,<stage>,<enable>,<lower_fr>,<upper_fr>,<state>	Response OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are saved automatically.

Parameter

<obj>	String type. "ant_cfg" Antenna tuning configuration file.
<stage>	Integer type. Antenna characterization stage. Range: 1–56.
<enable>	Integer type. 0 Enable corresponding stage 1 Disable corresponding stage
<lower_fr>	Integer type. Lower frequency of corresponding bandwidth, given in MHz multiplied by 10. Range: 4000–38000.
<upper_fr>	Integer type. Upper frequency of corresponding bandwidth, given in MHz multiplied by 10. Range: 4000–38000.
<state>	Integer type. Truth table combination of two different states of GRFC1 and GRFC2 pins respectively. Range: 0–3.
<err>	Integer type. Error code. See Chapter 3 for more information.

Example

```
AT+QCFG="setsyscfg","ant_cfg",1,1,8031,9600,2 // 803.1-960MHz
OK
```

NOTE

Overlapping between frequency ranges is forbidden.

2.1.2.18. AT+QCFG="uartmapping" Show Port and Function Mapping

The command is used to present port (physical and virtual) mapping, i.e., to present the function performed by each port.

AT+QCFG="uartmapping" Show Port and Function Mapping

Write Command
AT+QCFG="uartmapping"

Response
<port>:<function>,
...
<port>:<function>

OK

If there is an error related to ME functionality:
+CME ERROR: <err>

If there is any other error:
ERROR

Maximum Response Time

500 ms

Characteristics

The command takes effect immediately.

Parameter

<port>	String type.
DBG	Physical port, DBG UART
MAIN	Physical port, MAIN UART
AUX	Physical port, AUX UART
MCU	Internal port, Reserved (BG772A-GL, BG952A-GL)
EMUX0	Virtual port 0, EMUX function
EMUX1	Virtual port 1, EMUX function
EMUX2	Virtual port 2, EMUX function

	EMUX3	Virtual port 3, EMUX function
	ACM0	Virtual port 0, USB interface
	ACM1	Virtual port 1, USB interface
	ACM2	Virtual port 2, USB interface
	ACM3	Virtual port 3, USB interface
<function>	String type.	
	AT_PPP	AT client 0 / PPP data transfer
	AT_1	AT client 1
	AT_2	AT client 2
	GNSS_NMEA	GNSS NMEA output
	SFP_LOGGER	FW log messages, Debug mode
	FW_LOG	LTE log messages, Debug mode
	MAC_VIA_MAP	LTE log messages, Debug mode
	ALTCOM	Internal function, Reserved (BG772A-GL, BG952A-GL)
	None	No function is allocated to <port>
<err>	Integer type. Error code. See Chapter 3 for more information.	

2.1.2.19. AT+QCFG="uartmapping/profile" Read/Write Port Profile Mapping Configuration

This command reads or writes the value of parameters in port profile mapping configurations. The current mapping configuration of the selected port profile can be queried with **AT+QCFG="uartmapping"**.

AT+QCFG="uartmapping/profile" Read/Write Port Profile Mapping Configuration	
Write Command AT+QCFG="uartmapping/profile"[,<profile>]	Response If the optional parameter is omitted, query the current configuration: +QCFG: "uartmapping/profile",<profile> OK If the optional parameter is specified, set the port profile mapping configuration OK If there is an error related to ME functionality: ERROR If there is any other error:
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are saved automatically.

Parameter

<profile>	String type.	Delta change in relation to Default port profile
0	Default port profile	Mode USB+DEBUG DBG None MAIN AT1 AUX None ACM0 AT/PPP ACM3 GNSS
1	Port profile #1	Mode USB+DEBUG DBG GNSS_NMEA MAIN AT/PPP AUX AT1 ACM0 None ACM3 None

2.1.2.20. AT+QCFG="usb" Activate/Deactivate USB Interface

This command activates/deactivates USB Interface. To use USB interface, the module should be connected via USB cable to the host computer. Following the boot process, the module exposes 3 other ports.

AT+QCFG="usb" Activate/Deactivate USB Interface	
Write Command AT+QCFG="usb",<enable>	Response If the optional parameter is omitted, query the current settings: +QCFG: "usb",<enable> OK If the optional parameter is specified, activate/deactivate USB interface: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	1500 ms
Characteristics	The command takes effect after the module reboots.

The configuration is saved automatically.

Parameter

<enable>	Integer type. Activates/deactivates USB Interface. 0 Deactivate 1 Activate
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.2.21. AT+QCFG="usb/urcport" Configure URC Output Port in USB Interface Mode

The command queries and configures the URC output port in USB interface mode.

AT+QCFG="usb/urcport" Configure URC Output Port in USB Interface Mode	
Write Command AT+QCFG="usb/urcport"[,<URC_port>]	Response If the optional parameter is omitted, query the current settings: +QCFG: "usb/urcport",<URC_port> OK If the optional parameter is specified, configure the URC output port in USB Interface mode: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<URC_port>	Integer type. URC output port in USB Interface mode. 0 All ports 1 Virtual port 1 2 Virtual port 2 3 Virtual port 3
-------------------------	---

<err> Integer type. Error code. See **Chapter 3** for more information.

2.1.2.22. AT+QCFG="lte/bandprior" Configure Band Scan Priority under LTE

The command queries and configures the band scan priority.

AT+QCFG="lte/bandprior" Configure Band Scan Priority under LTE	
Write Command AT+QCFG="lte/bandprior" [<band_value>]	Response If the optional parameter is omitted, query the current settings: +QCFG: "lte/bandprior",<band_value> OK If the optional parameter is specified, configure the URC output port in USB Interface mode: OK If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The configuration is saved automatically.

Parameter

< band_value >	LTE Cat.M1 band value
01	Band 1
02	Band 2
03	Band 3
04	Band 4
05	Band 5
07	Band 7
08	Band 8

Example

```

AT+QCFG="lte/bandprior"           //Configure Band Scan Priority under LTE
+QCFG="lte/bandprior",1,5,7.....//Configured Band1, Band5 and Band7.
OK                                //Search priority: 1 -> 5 -> 7

```

2.1.2.23. AT+QCFG="stored/cell" Delete Stored Cell Acquisition DB

The command deletes stored cell acquisition DB

AT+QCFG="stored/cell" Delete Stored Cell Acquisition DB	
Write Command AT+QCFG="stored/cell"[,< value>]	Response If the optional parameter is omitted, query the current settings: +QCFG: "stored/cell",<value> OK If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The configuration is saved automatically.

Parameter

<value>	Integer value. 1 Delete acquisition DB
----------------------	---

3 Summary of +CME ERROR Codes

Final result code **+CME ERROR: <err>** indicates an error related to mobile equipment or network. The operation is similar to the **ERROR** result code.

<err> values are mostly used by common message commands. The following table lists most of general and GPRS related **ERROR** codes. For some GSM protocol failure causes described in GSM specifications, the corresponding **ERROR** codes are not included.

Table 2: Summary of +CME ERROR Codes

Numeric <err> Value	Verbose <err> Value
0	phone failure
1	no connection to phone
2	phone-adaptor link reserved
3	operation not allowed
4	operation not supported
5	PH-SIM PIN required
6	PH-FSIM PIN required
7	PH-FSIM PUK required
10	SIM not inserted
11	SIM PIN required
12	SIM PUK required
13	SIM failure
14	SIM busy
15	SIM wrong

16	incorrect password
17	SIM PIN2 required
18	SIM PUK2 required
20	memory full
21	invalid index
22	not found
23	memory failure
24	text string too long
25	invalid characters in text string
26	dial string too long
27	invalid characters in dial string
30	no network service
31	network timeout
32	network not allowed - emergency calls only
40	network personalization PIN required
41	network personalization PUK required
42	network subset personalization PIN required
43	network subset personalization PUK required
44	service provider personalization PIN required
45	service provider personalization PUK required
46	corporate personalization PIN required
47	corporate personalization PUK required

4 Appendix References

Table 3: Related Document

Document Name
[1] Quectel_BG77xA-GL&BG95xA-GL_AT_Commands_Manual

Table 4: Terms and Abbreviations

Abbreviation	Description
AP	Application Processor
APN	Access Point Name
CDS	Common Data Service
CE	Coverage Enhancement
eDRX	Extended Discontinuous Reception
EMM	EPS Mobility Management
EMUX	Multiplexing
eMTC	Enhanced Machine-Type Communication
GPIO	General-Purpose Input/Output
GSM	Global System for Mobile Communications
IMSI	International Mobile Subscriber Identity
JSON	JavaScript Object Notation
LED	Light Emitting Diode
LTE	Long-Term Evolution
LwM2M	Lightweight M2M

ME	Mobile Equipment
NB-IoT	Narrowband Internet of Things
PIN	Personal Identification Number
PS	Packet Switched
PSM	Power Saving Mode
PUK	PIN Unlock Key
RAT	Radio Access Technology
RRC	Radio Resource Control
SMS	Short Message Service
TA	Terminal Adapter
TAU	Tracking Area Update
URC	Unsolicited Result Code
(U)SIM	(Universal) Subscriber Identity Module